5 Solutions to Power Question

1. **a.** No. This equation would mean that $\min(\mathcal{O}, x) = x$ for all x. However, if $\mathcal{O}$ exists, then $\min(\mathcal{O}, \mathcal{O}+1) = \mathcal{O} \neq \mathcal{O}+1$. This is a contradiction, so no such $\mathcal{O}$ exists.

b. $\mathcal{I} = 0$. This makes $\mathcal{I} \odot x = \mathcal{I} + x = 0 + x = x$ for all real numbers x.

2. Translating this into "normal" arithmetic, the left-hand side is $a + \min(b, c)$, and the right-hand side is $\min(a+b, a+c)$. If $b \leq c$, then both sides are equal to $a+b$, and if $b > c$, then both sides are equal to $a+c$. Thus this equation holds for all real numbers a, b, c.

3. **a.** The equation translates to $\min(2x, x+2, 1) = \min(x, 2)$. If $x < \dfrac{1}{2}$, the equation is equivalent to $2x = x$, which implies $x = 0$. If $x \geq \dfrac{1}{2}$, the equation is equivalent to $1 = \min(x, 2)$, which implies $x = 1$. These both satisfy the original equation, and there cannot be any other solutions. Thus the two solutions are $x = 0$ and $x = 1$.

b. This equation translates to $\min(2x, -1) = \min(x, 1) + \min(x, -1)$. If $x \leq -1$, then both sides are equal to $2x$, so every $x \leq -1$ is a solution. If $-1 < x < -\frac{1}{2}$, then the equation becomes $2x = x - 1$, which has no solution on the interval. If $-\frac{1}{2} \leq x \leq 1$, then the equation is $-1 = x - 1$, so $x = 0$ is a solution. Finally, if $x > 1$, then the equation becomes $-1 = 0$, which is not true for any value of x. Thus the solutions to this equation are $x = 0$ and all $x \leq -1$.

4. Note that, translating into normal arithmetic, $a^{\odot b} = b \cdot a$. Each of the statements will be proven by translation.

a. This equation translates to $n(x+y) = nx + ny$, which is just the normal distributive property of multiplication over addition.

b. This translates to $n \cdot \min(x, y) = \min(nx, ny)$. If $x \leq y$, then both sides are equal to nx, and if $x > y$, then both sides are equal to ny. Either way, the equation holds for all real x and y and all integers n.

c. The first equality holds by part (b). For the second, the left-hand side is $\min(2x, 2y)$, and the right-hand side is $\min(2x, x+y, 2y)$. But note that $x+y$ is the arithmetic mean of $2x$ and $2y$, so it must be greater than or equal to one of them and less than or equal to the other. Thus $\min(2x, x+y, 2y) = \min(2x, 2y)$, as desired.

5. **a.** This equation translates to $2x = a$, which has exactly one solution: $x = \sqrt[\circ]{a} = \frac{a}{2}$.

b. This equation translates to $a + x = \mathcal{I} = 0$, which has exactly one solution: $x = \widehat{a} = -a$.

c. As shown in parts (a) and (b), the left-hand side is equal to $\frac{1}{2}(-a)$, and the right-hand side is equal to $-\left(\frac{1}{2}a\right)$, and these are equal by associativity of (normal) multiplication.

6. **a.** Let $f(x) = (0 \odot x^{\odot 2}) \oplus (0 \odot x) \oplus 1$. Then $f(x) = \min(2x, x, 1)$, which is $2x$ for $x \leq 0$, x for $0 < x < 1$, and 1 for $x \geq 1$. The only root of this PUP is $x = 0$.

b. The answer is yes. As an example, let $f(x) = (4 \odot x^{\odot 2}) \oplus (3 \odot x) \oplus (-1)$. Then $f(x) = \min(2x+4, x+3, -1)$, so $f(x) \leq -1$ for all x, which means f has no roots.

c. The answer is yes. As an example, let $f(x) = (0 \odot x^{\odot 2}) \oplus (0 \odot x) \oplus 0$. Then $f(x) = \min(2x, x, 0) = 0$ for all $x \geq 0$, so all nonnegative real numbers are roots of f.

Note: A general quadratic PUP $(a \odot x^{\odot 2}) \oplus (b \odot x) \oplus c$ has no roots if $c < 0$, infinitely many roots if $c = 0$, and one root if $c > 0$. A proof of this fact is in the solution to Problem 7.

d. Consider a general PUP, $f(x)$, as defined in the background to Problem 6. Then

$$f(x) = \min(nx + a_n, (n-1)x + a_{n-1}, \ldots, 2x + a_2, x + a_1, a_0).$$

This is the minimum of $n+1$ functions, all of whose graphs are lines of nonnegative slope. Suppose $x \leq y$. Then $kx + a_k \leq ky + a_k$ for each $k \geq 0$, and therefore $f(x) \leq f(y)$. Now suppose f has two different roots, so there are real numbers r_1, r_2 with $r_1 < r_2$ and $f(r_1) = f(r_2) = 0$. Now, for any z with $r_1 < z < r_2$, $f(r_1) \leq f(z) \leq f(r_2)$, so it follows that $f(z) = 0$. Thus every z in the interval $[r_1, r_2]$ is a root of f,

hence f has infinitely many roots. As seen in parts (a) and (b), if f does not have infinitely many roots, then it can only have either one root or zero roots. This shows that f has either zero, one, or infinitely many roots.

7. *There are multiple ways to solve this problem; in particular, the root can be found by translating to normal arithmetic. However, a solution using almost exclusively parallel-universe arithmetic follows.*

Setting $f(x) = 0$ and translating to normal arithmetic, note that the given equation is equivalent to $f(x) = \min(2x + a, x + b, c) = 0$. First, note that if $x > \max\left(\frac{c-a}{2}, c - b\right)$, then $f(x) = c$, and $f(x) \leq c$ for all x. Thus if $c < 0$, then f has no roots, and if $c = 0$, then f has infinitely many roots. Hence if f has exactly one root, it follows that $c > 0$. Notice that in general, $\widehat{(a^{\odot n})} = (\widehat{a})^{\odot n}$. This is because this equation translates to $-(na) = n(-a)$, which holds by associativity of (normal) multiplication. Now let $x = \widehat{\sqrt[\circ]{a} \oplus b}$; this is the root of f. Substituting and using the result from the above paragraph yields

$$f(x) = a \odot \widehat{\left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2}} \oplus b \odot \widehat{\left(\sqrt[\circ]{a} \oplus b\right)} \oplus c.$$

Form the parallel-universe equivalent of a common denominator:

$$f(x) = \left(a \odot \widehat{\left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2}}\right) \oplus \left(b \odot \left(\sqrt[\circ]{a} \oplus b\right) \odot \widehat{\left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2}}\right) \oplus \left(c \odot \left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2} \odot \widehat{\left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2}}\right).$$

Now use the distributive property, effectively combining like terms:

$$f(x) = \left[a \oplus b \odot \left(\sqrt[\circ]{a} \oplus b\right) \oplus c \odot \left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2}\right] \odot \widehat{\left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2}}.$$

Using the results of Problems 2 and 4 yields:

$$f(x) = \left[a \oplus \left(b \odot \sqrt[\circ]{a}\right) \oplus b^{\odot 2} \oplus (c \odot a) \oplus \left(c \odot b^{\odot 2}\right)\right] \odot \widehat{\left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2}}.$$

Now recall the assumption that $c > 0$, and therefore $a \oplus (c \odot a) = a$, and $b^{\odot 2} \oplus \left(c \odot b^{\odot 2}\right) = b^{\odot 2}$. Thus

$$f(x) = \left[a \oplus \left(b \odot \sqrt[\circ]{a}\right) \oplus b^{\odot 2}\right] \odot \widehat{\left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2}}.$$

By Problem 4c, this becomes:

$$f(x) = \left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2} \odot \widehat{\left(\sqrt[\circ]{a} \oplus b\right)^{\odot 2}} = 0,$$

as desired.

Other forms of the answer, such as $\sqrt[\circ]{\widehat{(a \oplus b^{\odot 2})}}$*, are also possible. However, answers like* $\sqrt[\circ]{\widehat{a}} \oplus \widehat{b}$ *are* ***not*** *valid, because Problem 4a does not hold if n is a negative integer.*

8. Before proving any part of this problem, let (k, r, s) be a factorization of f, and let

$$g(x) = k \odot (x \oplus r) \odot (x \oplus s),$$

so that $f(x) = g(x)$ for all x. By the result of Problem 2,

$$g(x) = (k \odot x^{\odot 2}) \oplus (k \odot (r \oplus s) \odot x) \oplus (k \odot r \odot s).$$

Also, note as in Problem 7 that $f(x) = \min(2x + a, x + b, c)$.

a. If $x < \min\left(\frac{c-a}{2}, b - a\right)$, then $f(x) = 2x + a = a \odot x^{\odot 2}$. Similarly, if $x < r$, then $g(x) = k \odot x^{\odot 2}$. Choose a value of x small enough to satisfy both inequalities. Because $f(x) = g(x)$, it follows that $a \odot x^{\odot 2} = k \odot x^{\odot 2}$. Multiplying (in the parallel universe) by $\widehat{(x^{\odot 2})}$ yields $k = a$, as desired.

b. As argued in Problem 7, if $x > \max\left(\frac{c-a}{2}, c - b\right)$, then $f(x) = c$. Similarly, if $x > s$, then $g(x) = k \odot r \odot s$. Choose x large enough to satisfy both inequalities. Because $f(x) = g(x)$, it follows that $c = k \odot r \odot s$. By part (a), $k = a$, and then by the definition of the parallel reciprocal, $r \odot s = c \odot \widehat{a}$.

9. As in Problem 8, let (k, r, s) be a factorization of a quadratic PUP $f(x)$, and let $g(x) = k \odot (x \oplus r) \odot (x \oplus s)$. Note by the distributive property (Problem 2), that

$$g(x) = (k \odot x^{\odot 2}) \oplus (k \odot r \odot x) \oplus (k \odot r \odot s)$$

because $r \leq s$ and thus $r \oplus s = r$. Furthermore, from Problems 8a and 8b, this becomes

$$g(x) = (a \odot x^{\odot 2}) \oplus (a \odot r \odot x) \oplus c.$$

Claim: If $2b < a + c$, then the only factorization of f is $a \odot \big(x \oplus (b - a)\big) \odot \big(x \oplus (c - b)\big)$ (where "$-$" denotes normal subtraction), and if $2b \geq a + c$, then the only factorization of f is $a \odot (x \oplus d) \odot (x \oplus d)$, where $d = \frac{c-a}{2}$.

To prove the claim, consider each case of the claim separately:

Case 1: Assume $2b < a + c$, so that $b - a < c - b$. Let $x = \frac{c-a}{2}$. Then

$$f(x) = \min(2x + a, x + b, c) = \min\left(c, \frac{c - a}{2} + b, c\right).$$

Because $2b < a + c$, it follows that $c - a < 2c - 2b$, and thus $x < c - b$. Then $x + b < c$, so $f(x) = x + b$.

Note also that

$$g(x) = \min(2x + a, x + a + r, c) = \min\left(c, \frac{c + a}{2} + r, c\right).$$

Because $f(x) = x + b < c$, conclude that $g(x) \neq c$, hence $g(x) = \frac{c+a}{2} + r$. Setting this equal to $f(x)$ yields $r = b - a$, and the result of Problem 8b gives $s = c - b$.

This is the only possible factorization of f, and it remains to show that it is a factorization. Recall that

$$g(x) = (a \odot x^{\odot 2}) \oplus (a \odot r \odot x) \oplus (a \odot r \odot s).$$

Using the definition of $\odot$, $a \odot r = b$ and $a \odot r \odot s = c$, so $g(x) = f(x)$ for all x, as desired.

Case 2: Now assume $2b \geq a + c$, so that $c - b \leq b - a$. First, note that for every x, either $x \leq b - a$ (and then $2x + a \leq x + b$), or $c - b \leq x$ (and then $c \leq x + b$). Thus

$$f(x) = \min(2x + a, x + b, c) = \min(2x + a, c).$$

Now consider the claim that $r \geq c - a - r$. If not, then there exists an x with $r < x < c - a - r$. Because $r < x$, it follows that $x + a + r < 2x + a$. Note that $x < c - a - r$ is equivalent to $x + a + r < c$. This implies that $g(x) = x + a + r$. And because $f(x)$ is equal to either $2x + a$ or c, it follows that $x = r$ or $x = c - a - r$, a contradiction. Therefore $r \geq c - a - r$, and rearranging, $r \geq \frac{c-a}{2}$.

Now consider s. By Problem 8b, $r + s = c - a$, and $r \leq s$ by assumption, so $2r \leq c - a$, and $r \leq \frac{c-a}{2}$. Thus the only possible factorization of f has $r = s = \frac{c-a}{2}$ (and $k = a$).

It remains to verify that this is a factorization of f. For convenience, let $d = \frac{c-a}{2}$. An earlier result showed that

$$f(x) = \min(2x + a, c) = (a \odot x^{\odot 2}) \oplus c.$$

Now apply the result of Problem 4b to g:

$$g(x) = a \odot (x \oplus d)^{\odot 2} = (a \odot x^{\odot 2}) \oplus (a \odot d^{\odot 2}) = (a \odot x^{\odot 2}) \oplus c = f(x),$$

as desired.

Authors' Note: Elle has in fact discovered the relatively new field of *tropical algebra*, which is currently an area of active mathematical research. For a further introduction to the field, including applications to algebraic geometry and computational biology, see the article "Tropical Mathematics" by David Speyer and Bernd Sturmfels, appearing in the June 2009 issue of *Mathematics Magazine*. This article was also invaluable to the authors as a source of inspiration and a few problems in this Power Question. It is available online at `https://math.berkeley.edu/~bernd/mathmag.pdf`.